Probabilistic Prediction of Outcome and Herding Time in Single-Pursuer Multi-Evader Herding

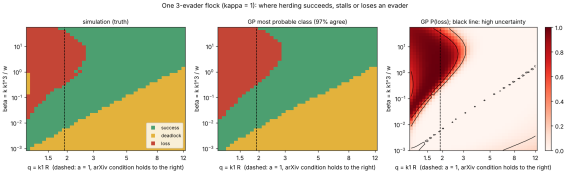
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EE604T Pattern Recognition and Machine Learning, IIT Dharwad

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The problem in one picture

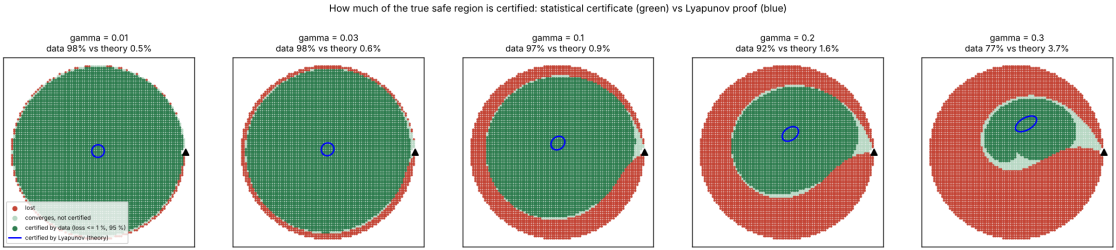
- One **pursuer** circles a goal; N **evaders** run away from it (speed k/d^2).
- The circle's radius follows the evader that was farthest at the start (arXiv 2505.10048).
- Three things can happen:
  - **success**: all evaders gathered on a small ring
  - **deadlock**: not settled within 300 laps
  - **loss**: an evader escapes
- **Question:** from the start alone, predict the outcome, *how sure* we are, and *how long* herding takes.



Why machine learning? The theory is local

- The papers prove a stable herd *near the final ring* if $a = 2(k_1 R)^2 e^{-2k_1\kappa} > 1$.
- They say nothing about *which starts* reach it, nor about time.
- In our data the theorem rule is a poor predictor: **45.7 %** balanced accuracy.
  - 52 % of successful runs violate $a > 1$; 11 % of lost runs satisfy it.
- For the circle law, the paper's Lyapunov ellipse certifies only **0.5–3.7 %** of the starts that really converge.

Headline: the proof certifies 0.5–3.7 %, the data certifies 77–98 %



- Data certificate: loss rate $\leq 1\,\%$ with $95\,\%$ confidence (exact Clopper–Pearson bound on unseen runs).
- arXiv law, test set: certified set **0.66 % loss, 91 % of successes**; the theorem's set $a > 1$: 8.8 % loss, 48 %.
- Honest: a guarantee on rates, not on every start (above 1 % at $\gamma = 0.3$ and for $N = 2$).

A readable rule: a conjecture for the theory

- Best 3-number rule for "an evader is lost" (86 % vs 67 % for the theorem's $a \leq 1$):

$$0.64 \log \beta - 1.48 \log a + 4.44\, \rho_{\max} > 3.35$$

- The theorem's own a appears, corrected for the push strength β and an evader near the edge ($\rho_{\max}$).
- Two numbers: lost if $q/\kappa < 2.9\, \beta^{0.22}$ (82 %).

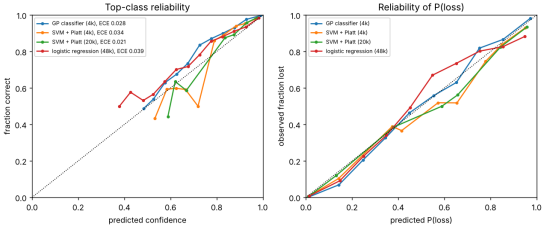
Data: 70,000 checked simulations

- Own simulator; checked against both papers' numbers and an accurate solver: **700/700** identical labels.
- Only two numbers matter after scaling: $q = k_1 R$ and $\beta = k k_1^3 / \omega$ (plus the start).
- 60,000 arXiv-law runs, $N = 1 \ldots 6$, three strata of a, Latin hypercube; 10,000 circle-law runs.
- Labels: 26,454 success, 16,013 deadlock, 17,533 loss.
- **Features** (24): describe only the start; do not depend on the evaders' order; work for any N.

Classifiers: accuracy and honest probabilities

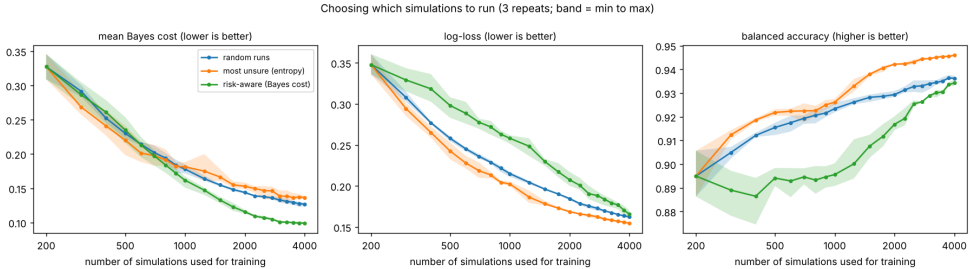
model (train runs)	bal. acc.	log-loss
theory rule	0.457	–
logistic regression (48k)	0.825	0.482
SVM + Platt (20k)	0.938	0.200
GP classifier (4k)	**0.937**	**0.162**

- Train $N \leq 4$, test $N = 5, 6$: **92.8 %**.
- GP answers its 80 % most confident runs: **98.6 %**.
- Bayes rule (losing an evader costs most): "loss called success" 169 $\rightarrow$ **9**.



Novelty 1: risk-aware active learning

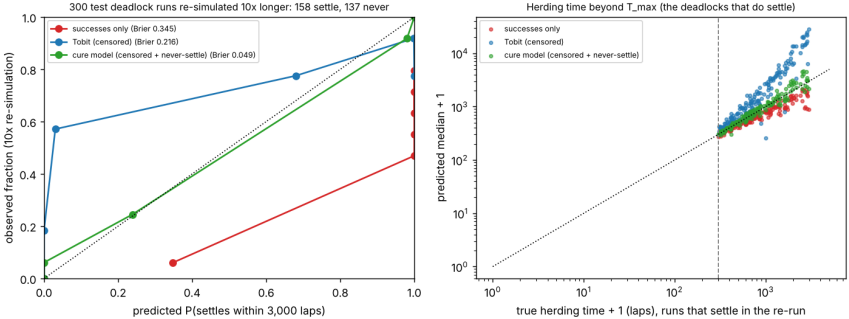
- Simulations cost time: let the GP **choose the next simulation**.
- *Entropy*: where the model is unsure. *Risk-aware* (ours): where a wrong decision costs most.



- Risk-aware: same decision cost as random with **2.3**× **fewer** runs; entropy: best accuracy but *worse* decisions.

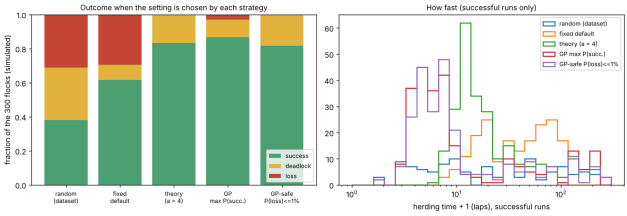
Novelty 2: herding time with censoring (slow vs stuck)

- Deadlock runs only tell us "time $>$ 300 laps". Dropping them biases the model.
- Kernel **mixture-cure** survival model: P(ever settles) + time if it settles.
- Check: 300 deadlock runs re-simulated **10$\times$ longer**: 158 settle, 137 never. From the start alone the cure model predicts which: **AUC 0.98**; mean prediction 53.5 % vs observed 53.6 %.

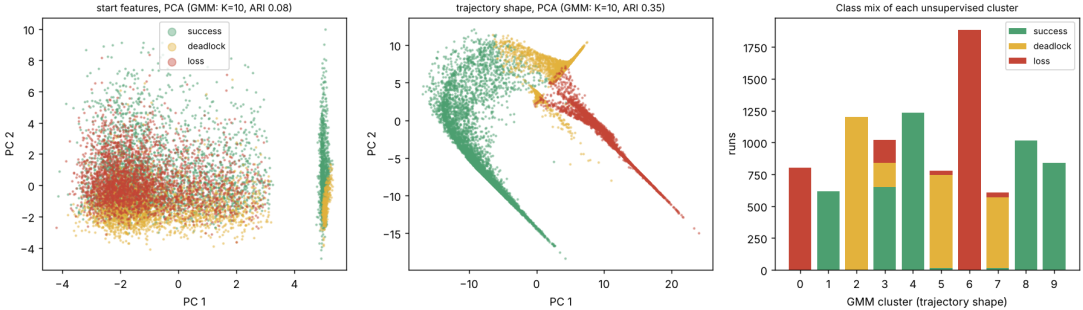


Novelty 3: using the model to choose the pursuer settings

- 300 flocks; each strategy picks q and β; **every choice simulated**.
- Theory rule ($a = 4$): 83 % success, **0 % loss**.
- GP max P(success): 87 % but 3 % loss.
- **GP-safe** (P(loss) $\leq 1\,\%$): 0 % loss and **2.3× faster** (6 vs 14 laps, Wilcoxon $p \approx 10^{-23}$).
- Lesson again: put the cost of the worst mistake into the decision.



Do the regimes appear without labels? (GMM + PCA)



- From the start: no (ARI 0.08) $\Rightarrow$ supervision is needed.
- From the trajectory shape: three arms, clusters **95 % pure** $\Rightarrow$ the labels are natural regimes.

Set kernel: compare flocks as sets of points

- Instead of hand-made summaries, compare two flocks evader by evader:

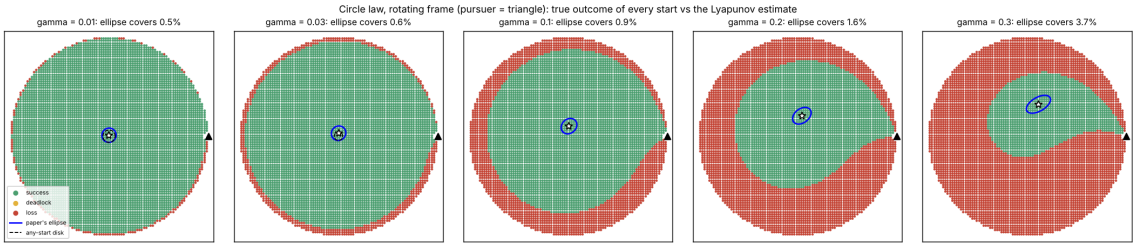
$$K(A,B) = k_g(g_A, g_B)\Big[k_e(e_{0A}, e_{0B}) + \tfrac{1}{|S_A||S_B|}\sum_{u\in S_A, v\in S_B} k_e(u,v)\Big]$$

- Works for any number of evaders (kernel mean embedding).

SVM, balanced accuracy	random split	**train** $N \le 4$**, test** $N = 5, 6$
hand-made features	93.0 %	90.0 %
set kernel	93.6 %	**93.8 %**

- The set kernel does not lose accuracy on bigger flocks than it has seen.

Theory vs data: the circle law



- Blue: the paper's Lyapunov ellipse (always correct). Green: starts that really converge.
- The proof covers 0.5–3.7 % of the true region: near the ring the motion is an almost undamped rotation.
- The classifiers learn the true region: 99 % balanced accuracy for this law.

Honest limits

- Everything is simulation; labels depend on our numbers ($T_{\max} = 300$ laps, 1 %).
- Exact GP limited to 4,000 training runs ($O(n^3)$); kernel tuned on 1,000.
- The cost matrix is a design choice.
- Active learning chooses from a pool of already-simulated runs (equivalent here: the simulator is deterministic).

Summary

- Calibrated, uncertainty-aware prediction of outcome and herding time, for any flock size.
- Statistical certificate: 77–98 % of the true safe region vs 0.5–3.7 % for the proof.
- Shows where the local theory stops protecting the evaders.
- Censored survival model separates slow from stuck herds.
- Risk-aware active learning: choose simulations by the decision you will make.
- Live demo: set up a flock $\rightarrow$ probabilities, decision, time $\rightarrow$ simulate and compare.